

Week 1.11 — Do the Math — Buoyancy Stops Being a Guess, and the Company Starts Acting Like One

Sunday, August 30, 2026 · 3:00 – 5:00 PM · 14 students

Fourteen of us at 1.11 — Trenton, Isaac, Luke, Maddox, Carter, Rebecca, Sam, Oliver, Eli, Rhylan, Tyler, Adrian, Lillian, and Annabelle — from 3:00 to 5:00, matching the season's biggest turnout. No pool this week and no build. This was a desk session, and it moved more than most build nights do. The headline is buoyancy. Back at 1.8 we wrote down that we can't tune buoyancy, that it's whatever the foam happens to give us. This week we stopped guessing and started calculating: we worked out the water displacement of the PVC pipe we have on hand, turned that number into how much lift each section actually provides, and designed a buoyancy solution around the arithmetic instead of around a hunch. That is the difference between a vehicle that floats and a vehicle you can trim. Alongside that, the mechanical track kept closing out the PVC frame model in CAD and produced a V1 design of the payload manipulator, with a cost breakdown of the whole concept so nobody falls in love with a claw we can't afford. The electrical track spent the session learning to read and draw schematics — a schematic is a language, and until you can read one you're guessing at what a circuit does. The sensor track ran real comparisons: three camera modules against each other and two pressure sensors against each other, judged against what the missions need rather than what looks impressive. Management worked the integrated master schedule and the budget, and started the conversation about the one thing a MATE company needs that we don't have yet — a name.

14	3	2	V1
STUDENTS — MATCHED THE SEASON HIGH	CAMERA MODULES COMPARED HEAD TO HEAD	PRESSURE SENSORS ON THE SHORT LIST	MANIPULATOR DESIGN, IN CAD AND COSTED

WHAT WE LEARNED

- Buoyancy is a calculation, not a feeling. The displacement of a length of PVC pipe is a number you can work out at a desk, before you cut anything, and it tells you exactly how much lift you're going to get.
- A schematic is a language. Until you can read one, you are guessing at what a circuit does — and a team that guesses at its own wiring cannot troubleshoot it under pressure.
- Researching a part is not reading reviews. It is writing down what the mission requires and then checking each candidate against that list. Three cameras compared against requirements beats one camera chosen because it looked good.
- An integrated master schedule that hasn't been integrated is just a document. The value is in the merge — where two sub-teams find out they both need the same week.
- Cost belongs in the design conversation at concept, not after the build. The manipulator got a cost sheet at V1, which is the right time to learn a design is too expensive.
- MATE teams compete as companies, and a company has a name. That's not decoration — it's what goes on the technical documentation, the marketing display, and the poster at the regional.

DECISIONS MADE

- Buoyancy is arithmetic from here forward. We calculated the water displacement of the PVC we have and built the design around that number — the 1.8 finding that we can't tune buoyancy is now being answered with math instead of more foam.
- The team needs a company name. Say Watt Robotics is the program; MATE teams compete as companies, and the company needs its own identity before the technical documentation and marketing display get written.
- The integrated master schedule stops being a document and gets integrated. Every sub-team's dates land on one calendar, and the schedule has to exist before the 2027 game manual drops in November.
- The budget gets finished against the grants actually in hand, not against a wish list. Cost enters the design conversation now, at concept, not after somebody has built the thing.
- Parts get compared against requirements before anything is bought. Cameras and pressure sensors go on a written short list, and the recommendation goes to the CFO with the reasoning attached.

ACTION ITEMS

Bring three candidate company names to the team for a vote

Trenton (CEO) · by 1.12

Integrate the master schedule — every sub-team's deliverables on one calendar, before November

Annabelle (COO) · by 1.12

Close the budget against confirmed grant funding

Maddox (CFO) · by 1.12

Finish the PVC frame model in CAD

Adrian (Mechanical) · by 1.12

Build the calculated buoyancy design and test whether the vehicle trims to neutral

Mechanical group · by 1.12

Pick a schematic and PCB drawing tool the whole electrical track will use

Electrical group · by 1.12

Narrow to one camera and one pressure sensor, with the comparison written down

Rhylan (Sensor) · by 1.12

Cost out the V1 manipulator concept and bring it to the CFO

Eli (Payload) · by 1.12

STUDENT LEARNING OUTCOMES

- Calculated the water displacement of PVC pipe and converted it into usable lift, then designed a buoyancy solution from that number.
- Learned to read and draw electrical schematics — the notation, what the symbols mean, and how to trace a circuit on paper.
- Advanced the PVC frame model in CAD toward a finished assembly.
- Produced a V1 CAD design of the payload manipulator and built a cost breakdown of the concept.
- Ran a structured parts comparison: three camera modules and two pressure sensors, judged against mission requirements.
- Worked the integrated master schedule and the grant-based budget — the management side of an engineering program, not just the build side.
- Started the outreach and marketing track that MATE scores separately from the vehicle.

LOOKING AHEAD

Class 1.12 closes Phase 1 — the final TriggerFish demonstration for parents, stakeholders, and sponsors, plus the role retrospective. Before then: the company name goes to a vote, the budget closes, the buoyancy design gets built and tested against the calculation, and the sensor track brings back one camera and one pressure sensor with the reasoning written down. Two items from the 1.11 plan did not happen this session and carry forward — capsule dive #3 (the gate is three consecutive dry dives) and the first walkthrough of a System Integration Diagram.

MADE POSSIBLE BY

